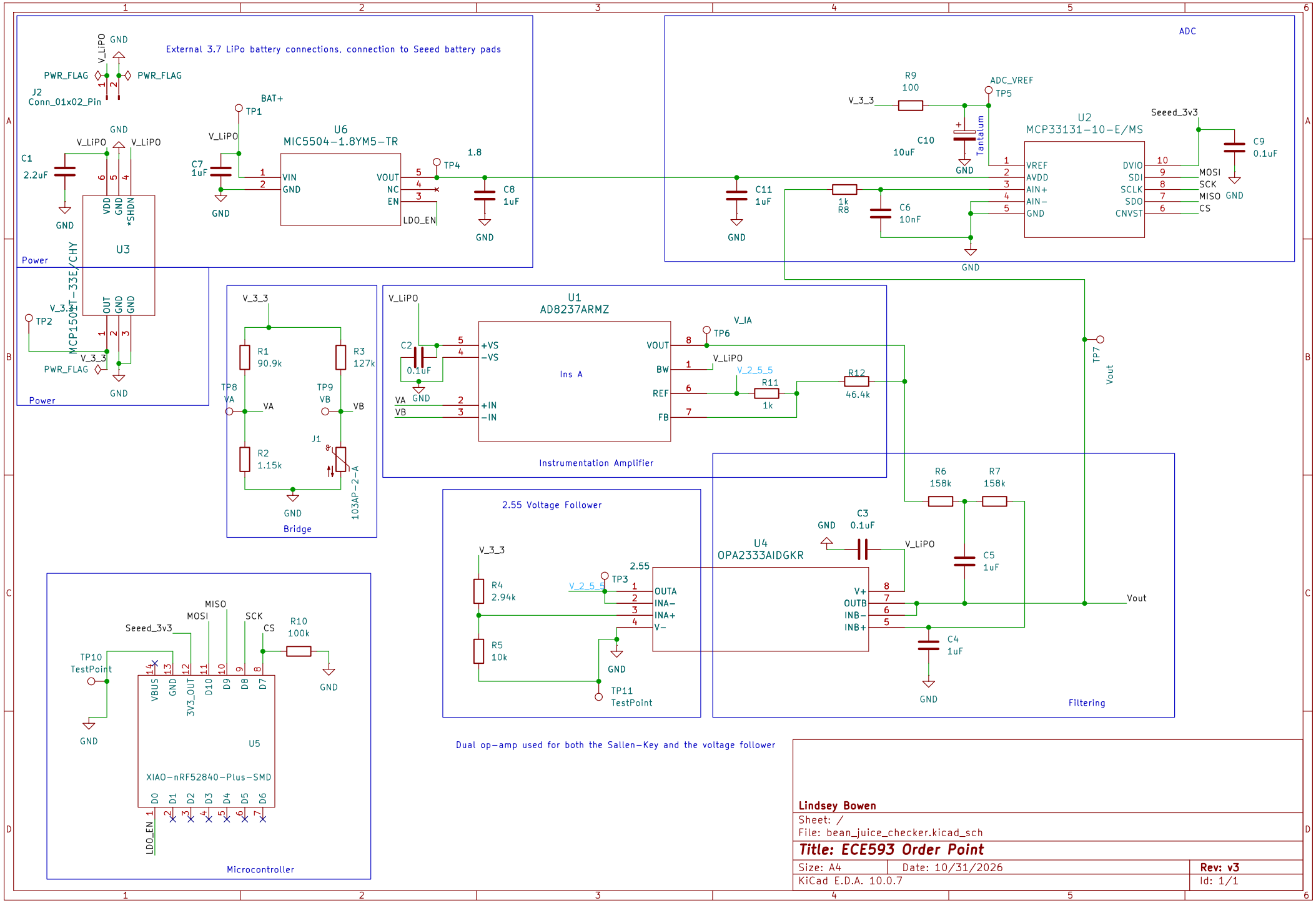

External 3.7 LiPo battery connections, connection to Seeed battery pads
J2
Conn_01x02_Pin
V_LiPO
GND
PWR_FLAG
U6
MIC5504-1.8YM5-TR
BAT+
TP1
TP4
1.8
C1
2.2uF
C7
1uF
C8
1uF
LDO_EN
U3
MCP1501T-33E/CHY
TP2
V_3_3
Power
ADC
U2
MCP33131-10-E/MS
R9
100
ADC_VREF
TP5
C10
10uF
Tantalum
C11
1uF
R8
1k
C6
10nF
C9
0.1uF
Seeed_3v3
MOSI
SCK
MISO
CS
U1
AD8237ARMZ
Ins A
C2
0.1uF
V_IA
TP6
R1
90.9k
R2
1.15k
R3
127k
R11
1k
R12
46.4k
TP7
Vout
Instrumentation Amplifier
Bridge
TP8
TP9
VA
VB
J1
103AP-2-A
2.55 Voltage Follower
R4
2.94k
R5
10k
TP3
2.55
TP11
TestPoint
U4
OPA2333AIDGKR
R6
158k
R7
158k
C3
0.1uF
C5
1uF
C4
1uF
Filtering
Dual op-amp used for both the Sallen-Key and the voltage follower
TP10
TestPoint
R10
100k
U5
XIAO-nRF52840-Plus-SMD
LDO_EN
Microcontroller
Lindsey Bowen
Sheet: /
File: bean_juice_checker.kicad_sch
Title: ECE593 Order Point
Size: A4
Date: 10/31/2026
Rev: v3
KiCad E.D.A. 10.0.7
Id: 1/1